Emerging Research Frontiers in NLP

Small Models, Diffusion LMs, Hallucination, Interpretability, and LLM Judges

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12. Summary

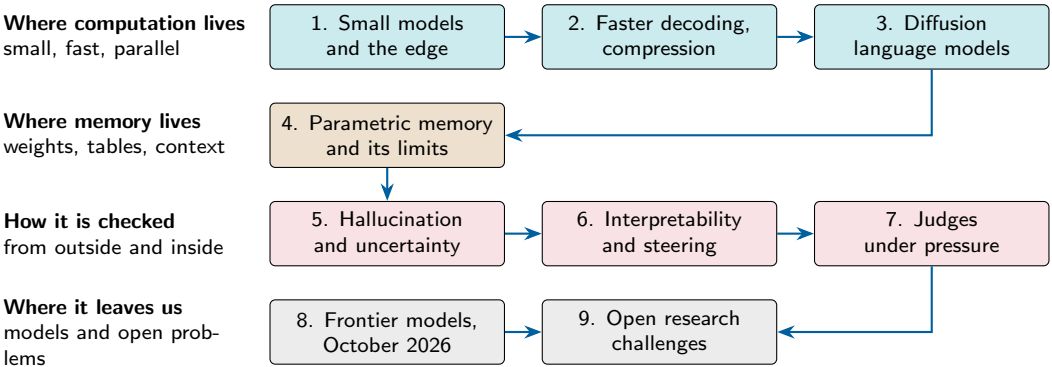
The Frontier Is No Longer a Bigger Dense Model

- Architectures, reasoning, post-training and agents all changed in 2025 and 2026. Today asks: **what is still moving?**
- A 2026 phone-sized model, Gemma 4 E2B, *"roughly matches Gemma 3 27B performance with 10x less parameters"*.
- So the research questions changed. Three of them organise this lecture:
  - **Where does computation live?** On the edge, in a drafter, in a parallel denoiser.
  - **Where does memory live?** In dense weights, in addressable tables, in the context.
  - **How is any of it checked?** By uncertainty estimates, by looking inside, by judges.

Ask for the honest number. Speculative decoding with EAGLE-3 gives 4.40x at batch size 1, and 1.38x at batch size 64, on the same model.

Gemma Team, "Gemma 4 Technical Report" (2026), arXiv:2607.02770. Li et al., "EAGLE-3," arXiv:2503.01840, Tables 1 and 3 (LLaMA-3.1-8B).

Today's Map: Nine Topics, One Story



Rows follow the three questions of the previous slide. The last row turns to the models of October 2026 and the problems they leave open.

Learning Outcomes

By the end of this session, you should be able to:

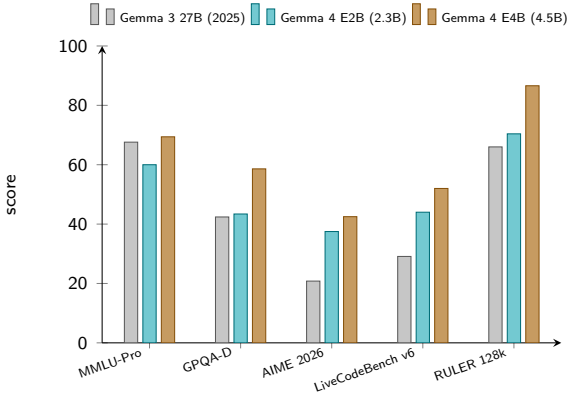
- **Explain** what makes a small model strong in 2026, and what limits it on a device.
- **Compare** drafter-free speculation, pruning and low-rank methods by *realised* speedup.
- **Describe** how diffusion LMs are built and sped up, and where they trail AR models.
- **Analyse** parametric memory: its capacity, and its freshness and attribution limits.
- **Explain** why stronger reasoners can hallucinate more, and choose an uncertainty estimator.
- **Evaluate** what probes, sparse autoencoders and steering vectors can and cannot show.
- **Assess** judges, reward models and leaderboards, which degrade when optimised.
- **Formulate** a research question from the October 2026 frontier.

Prerequisite knowledge: speculative decoding; KV caching, GQA and sliding-window attention; prompt compression and KV-cache eviction; quantization, pruning and distillation, including on-policy distillation; mixture-of-experts, delta-rule linear attention and Titans; masked diffusion LMs (LLaDA); GRPO; reward models, LLM-as-a-judge and Chatbot Arena; binary grading and guessing, semantic entropy, Chain-of-Verification, fine-tuning on unknown facts; the refusal direction; chain-of-thought monitoring; agent time horizons; scaling laws and synthetic data.

Research Frontiers: **Small Models and the Edge**

How much model fits in a phone, and what makes it good?

Ten Times Fewer Parameters, the Same Scores



- One generation later, a model with 2.3B effective parameters matches a 27B one.
- **Read it carefully.** The Gemma 4 models run with thinking. The Gemma 3 baseline does not.
- Size did not do this. Four levers did:
  - data and distillation
  - fewer bits, trained in
  - architecture shaped by memory
  - respect for edge hardware

Gemma Team, Google DeepMind, "Gemma 4 Technical Report" (v2, July 2026), arXiv:2607.02770, Tables 5 and 9.

What "Small" Means, and Who Builds It

- **Working definition:** fits on a consumer device and serves one user. Under about 10B parameters.

Model	Parameters	Training tokens	Distinctive idea
SmolLM3 (Jul 2025)	3B	11.2T	staged data mix
MobileLLM-R1 (Sep 2025)	0.95B	4.2T	data chosen by influence
Apple on-device (2025)	about 3B	14T	2-bit weights, shared KV cache
BitNet b1.58 (Apr 2025)	2B	4T	ternary weights, trained natively
LFM2 (Nov 2025)	0.35 to 2.6B	10T + 1T	convolutions and a little attention
Qwen3.5 Small (Mar 2026)	0.8 to 9B	not disclosed	3:1 Gated DeltaNet hybrid
Gemma 4 E2B, E4B (2026)	2.3B, 4.5B eff.	not disclosed	per-layer embeddings, KV sharing

- **Over-training is the norm:** thousands of tokens per parameter, against about 20 for Chinchilla.
- **A 2026 habit:** labs no longer publish token counts or distillation recipes.

Belcak et al., "Small Language Models are the Future of Agentic AI," arXiv:2506.02153. Technical reports and model cards of each model, including arXiv:2509.24945, arXiv:2507.13575, arXiv:2511.23404, arXiv:2607.02770.

Lever 1: Data and Distillation

Recipe	Where	What the report shows
Curation instead of volume	MobileLLM-R1	4.2T tokens, 11.7% of Qwen3's 36T. Wins AIME24 (15.5 against 11.3), loses MATH500 (26.8 against 29.8)
Sampled-logit distillation	Gemma 3	256 teacher logits sampled per token. A small teacher is better for short runs, a large one for long runs

- **LFM2 refines top-K distillation:** match the teacher's *total mass* on its top-32 tokens, and its *distribution inside* that set, as two terms: $\mathcal{L}_{DTK} = \mathcal{L}_B + \mathcal{L}_T$.
- Microsoft's 2026 verdict on its Phi small models: *"data quality remains the primary lever"*.

Meta, arXiv:2509.24945. Gemma 3, arXiv:2503.19786. Qwen3, arXiv:2505.09388. Liquid AI, arXiv:2511.23404. Aneja et al., arXiv:2603.03975.

Lever 2: Fewer Bits, Trained In

- The shift beyond post-training quantisation: **train at the target precision**. BitNet rounds every weight to $\{-1, 0, 1\}$ during training, with γ the mean absolute weight:

$$\tilde{W} = \mathrm{RoundClip}\Big(\frac{W}{\gamma + \epsilon}, -1, 1\Big)$$

Model	Memory	CPU decode	Energy per token	Benchmark average
LLaMA-3.2-1B	2.0 GB	48 ms	0.258 J	44.90
Qwen2.5-1.5B	2.6 GB	65 ms	0.347 J	55.23
BitNet b1.58 2B4T	0.4 GB	29 ms	0.028 J	54.19

- **Apple** ships 2 bits per weight (MMLU 67.8 to 64.4). **Gemma 4 E2B** goes from 4.6 GB to 0.8 GB.
- Note: BitNet's numbers need its own runtime, and its memory excludes embeddings.

Microsoft Research, "BitNet b1.58 2B4T Technical Report," arXiv:2504.12285, Table 1. Ma et al., arXiv:2402.17764. Apple, arXiv:2507.13575. Gemma 4, arXiv:2607.02770, Table 3.

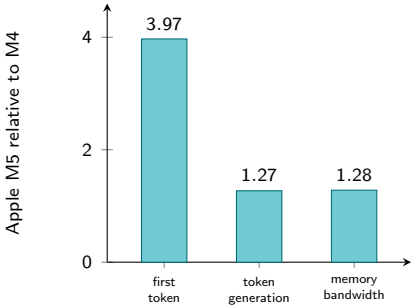
Lever 3: Architecture Shaped by the KV Cache

Design choice	Model	Effect
Later block reuses the earlier block's KV	Apple (2025)	KV memory and first-token time down 37.5%
Per-layer embeddings off the accelerator	Gemma 4 E2B	2,340M of about 5B parameters offloaded
Convolutions plus a few attention blocks	LFM2	found by search with phones in the loop

- **The shared goal:** fit weights, KV cache and buffers into a **memory budget of a few GB**.
- **An open disagreement.** LFM2's search found that linear-attention operators hurt on-device latency without improving quality. Qwen3.5 Small ships one anyway.

Apple, arXiv:2507.13575. Gemma 4, arXiv:2607.02770, Table 1. LFM2, arXiv:2511.23404. Qwen3.5-2B model card (2026).

Lever 4: Edge Physics



Token generation: 1.19x to 1.27x across models.

- **Prefill is compute-bound. Decode is bandwidth-bound.** A faster GPU gave 3.97x on the first token and about 1.2x after it, the same as the gain in memory bandwidth.
- **Memory is mostly not weights.** At maximum context, the KV cache and buffers take 83 to 90% of memory.
- **4 bits is the practical floor.** Irregular 3, 5 and 6-bit formats run slower on less optimised kernels.
- **NPUs are not a free win.** One 2026 abstract reports CPUs up to 1.6x faster at prefill.

Apple, "Exploring LLMs with MLX and the Neural Accelerators in the M5 GPU" (Nov 2025). Lu et al., "Small Language Models: Survey, Measurements, and Insights," arXiv:2409.15790. Li, Qi, Chen, arXiv:2605.27435 (abstract only).

Where Small Models Still Lose, and Where They May Win

Benchmark	Qwen 3.5-4B	Qwen 3.5-9B	GPT-OSS 120B
MMLU-Pro	79.1	82.5	80.8
GPQA Diamond	76.2	81.7	80.1
IFEval	89.8	91.5	88.9
HMMT Feb 25	74.0	83.2	90.0
LiveCodeBench v6	55.8	65.6	82.7

Thinking mode, from the Qwen3.5-9B model card (March 2026).

- **Knowledge and instructions:** a 9B model matches one more than ten times its size.
- **Hard maths and code:** the gap is still 7 to 17 points.
- **The agent argument (NVIDIA).** A 7B model costs 10 to 30 times less to serve than a 70 to 175B one. In three open agents, 40 to 70% of LLM calls could go to a small model.
- **The condition:** *fine-tuned* small models inside a mixed system, not a small model as the orchestrator.

Qwen team, Qwen3.5-9B model card (2026). Belcak et al., NVIDIA, "Small Language Models are the Future of Agentic AI," arXiv:2506.02153, Appendix B.

Small Models: What to Remember

- A 2026 model with **2 to 5B effective parameters** reaches the scores of a 27B model from 2025.
- The gain comes from **data curation, distillation and heavy over-training**, not from a new block.
- **Low precision is trained in**: ternary weights, 2-bit QAT, official int4 checkpoints.
- Edge architectures are designed around **KV memory and memory bandwidth**.
- Small models still trail on hard code and maths. Their strongest case is as **fine-tuned specialists**.

Next. If decode speed is set by memory bandwidth, the same holds for large models in the data centre. What is left to gain once the weights are already quantised?

Research Frontiers: **Efficiency Beyond Quantisation**

The weights are already 4-bit. Where does the next speedup come from?

Speculative Decoding Pays Off Only at Small Batch

- **Why it works.** Generating one token per sequence reads all the weights for one small matrix-vector product, so the GPU waits on memory and its compute sits idle. Verifying several drafted tokens in one pass uses that idle compute for free.
- **Why it stops working.** With many sequences in the batch, the compute is no longer idle. Verifying draft tokens now competes with real work, and the gain shrinks or turns into a loss.

Speed relative to plain decoding	batch 2	batch 16	batch 64
EAGLE-3	1.81x faster	1.48x faster	1.38x faster
EAGLE-2	1.40x faster	slower than plain decoding from batch 24 (0.88x to 0.99x)	

LLaMA-3.1-8B served with SGLang on one H100.

Speculative decoding is a **low-batch latency** tool. Under heavy load, vLLM measured 1.4x to 1.8x *slowdowns*.

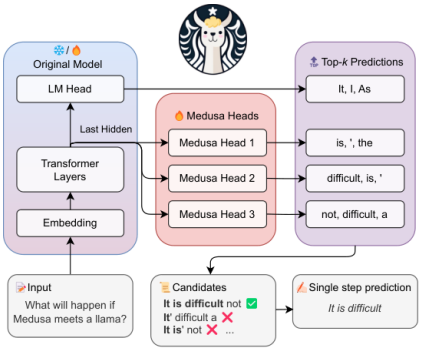
Liang, Stanford CS336 Lecture 10 (2025). Li et al., "EAGLE-3," arXiv:2503.01840, Table 3. vLLM Team, "How Speculative Decoding Boosts vLLM Performance by up to 2.8x" (Oct 2024).

Medusa: Let the Model Draft for Itself

- **Problem:** a separate draft model must be found, aligned and served.
- **Idea:** add K small heads on the last hidden state. Head k predicts the token $k+1$ steps ahead:

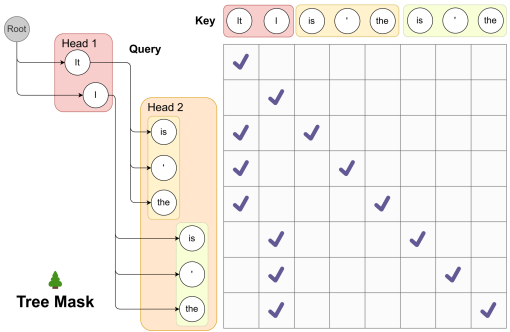
$$p_t^{(k)} = \mathrm{softmax}\big(W_2^{(k)}\,(\mathrm{SiLU}(W_1^{(k)} h_t) + h_t)\big)$$

- **Medusa-1:** heads on a frozen backbone, over 2.2x. **Medusa-2:** joint training, 2.3 to 3.6x.
- Each step costs 1.18 to 1.27x more, so speedup = tokens per step / overhead.



Cai, Li, Geng, Peng, Lee, Chen, Dao, "Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads," arXiv:2401.10774, Figure 1 and Table 1.

Verifying a Whole Tree in One Forward Pass



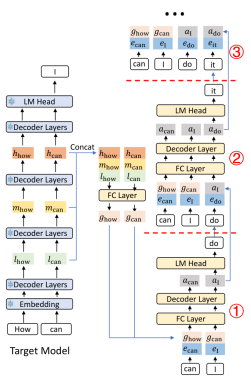
- The top guesses of each head combine into a **tree of candidates**.
- A **tree attention mask** lets each candidate see only its ancestors. One pass scores every branch.
- Bigger is not better: a sparse 64-node tree beats a dense 256-node one.
- **Hydra** lets each head see the tokens drafted before it. Up to 1.31x over Medusa.
- **Lossy or lossless?** Medusa's *typical acceptance* changes the output distribution. Standard verification does not.

Cai et al., arXiv:2401.10774, Figures 2 and 4. Ankner et al., "Hydra: Sequentially-Dependent Draft Heads for Medusa Decoding," arXiv:2402.05109.

EAGLE: Draft in Feature Space, Then Train as You Test

- **EAGLE (2024).** A one-layer drafter autoregresses over the target's *hidden features*, not its tokens.
- **EAGLE-2.** The drafter's confidence predicts acceptance, so the tree grows where acceptance is likely.
- **EAGLE-3 (2025).** Predict tokens directly, and feed the drafter *its own outputs* during training.

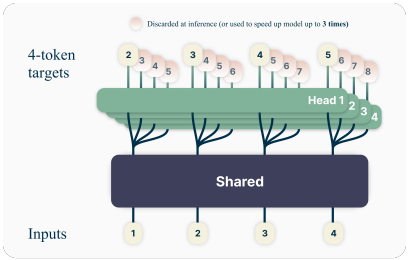
Vicuna-13B, MT-bench	Speedup	Tokens per step
Vanilla speculative sampling	1.93x	2.27
EAGLE-2	4.26x	4.83
EAGLE-3	5.58x	6.65



Li, Wei, Zhang, Zhang, "EAGLE," arXiv:2401.15077. "EAGLE-2," arXiv:2406.16858. "EAGLE-3," arXiv:2503.01840, Table 1 and method figure. Batch size 1. All three are lossless.

Multi-Token Prediction: the Drafter Ships with the Model

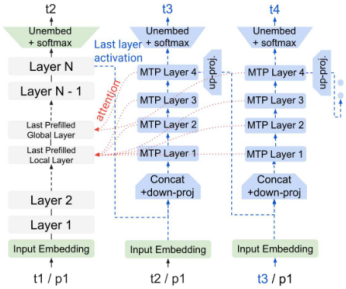
- **Idea:** train the model itself to predict several future tokens from a shared trunk. The extra heads are a free drafter.
- As a training loss it hurts below about 1B parameters and helps from about 3B.
- **DeepSeek-V3:** second-token acceptance of 85 to 90%, which gives **1.8x tokens per second**.
- **A train and serve mismatch.** Engines run one head recursively, but it was trained on true context. Acceptance by position on Qwen3-Next: 0.897, 0.719, 0.476.
- **FastMTP** retrains the head recursively: 0.912, 0.776, 0.616.



Gloeckle et al., "Better & Faster Large Language Models via Multi-token Prediction," arXiv:2404.19737, Figure 1. DeepSeek-AI, arXiv:2412.19437. Red Hat Developer, FastMTP article (Sep 2026).

2026: Drafters That Do Not Autoregress

- **DFlash (ICML 2026).** A small *block-diffusion* drafter writes a whole block in one pass, conditioned on the target's hidden features. Lossless, over 6x, up to 2.5 times EAGLE-3's speedup.
- **Saguaro.** While the target verifies, a drafter on separate hardware already drafts for the likely outcomes. About 30% faster than tuned speculative decoding.
- **Gemma 4** ships a 4-layer drafter that reads the main model's KV cache. Its output layer picks among 4,096 token clusters, not 262,000 tokens.



NVIDIA reports up to 15x for DFlash on gpt-oss-120b. That is a vendor blog, on its newest hardware.

Chen, Liang, Liu, "DFlash," arXiv:2602.06036. Kumar, Dao, May, "Speculative Speculative Decoding," arXiv:2603.03251. Gemma 4, arXiv:2607.02770, Figure 1. NVIDIA developer blog (June 2026).

Pruning Weights: "50% Sparse" Is Not "2x Faster"

- **Wanda** needs no retraining. It scores each weight by its size *times* the norm of the input it multiplies: $S_{ij} = |W_{ij}| \cdot \|X_j\|_2$.

WikiText perplexity at 50% sparsity	LLaMA-7B	13B	30B	65B
Magnitude pruning	17.29	20.21	7.54	5.90
SparseGPT (minutes of solving)	7.22	6.21	5.31	4.57
Wanda (seconds)	7.26	6.15	5.24	4.57

- **Does it run faster?** Unstructured zeros do not. The hardware-friendly 2:4 pattern gives about 1.6x on the matrix multiply alone.

- **End to end,** the first one-shot result on recent GPUs is 1.2x in decoding (SpenseGPT, 2026), and only by leaving some regions dense.

Sun, Liu, Bair, Kolter, "Wanda," arXiv:2306.11695, Tables 3 to 5. Frantar, Alistarh, "SparseGPT," arXiv:2301.00774. Lee, Hwang, Rajbhandari, "SpenseGPT," arXiv:2606.10445.

Structured Pruning Ships When Distillation Follows It

- Removing whole neurons or layers gives an ordinary dense model that any runtime can serve.
- **Read the headline twice.** LLM-Pruner's "keeps 94.97% of performance" is its best row (60.07 of 63.25). Layer removal keeps multiple-choice scores while summarisation nearly collapses.

Llama-3.1 8B to 4B (NVIDIA Minitron)	MMLU	Throughput	Training tokens
Original 8B	0.6528	1.0x	15T
Width-pruned 4B	0.6053	about 1.8x	94B
Depth-pruned 4B	0.5871	about 2.7x	94B

- **The recipe:** train the largest model once, prune by activation importance, retrain with distillation only. **Width keeps accuracy. Depth buys throughput.**

Ma, Fang, Wang, "LLM-Pruner," arXiv:2305.11627, Table 1. Men et al., "ShortGPT," arXiv:2403.03853. Muralidharan et al., arXiv:2407.14679. NVIDIA developer blog, Llama-3.1-Minitron 4B (Aug 2024).

Low-Rank Compression in One Equation

- Plain SVD of W fails: a small singular value can still matter for the *inputs the layer sees*.
- **SVD-LLM** whitens first. With $SS^\top = XX^\top$, decompose $WS = U\Sigma V^\top$ and truncate. Each dropped singular value then costs exactly its own size in output error:

$$L = \|WX - W'X\|_F, \qquad L_i = \sigma_i$$

LLaMA-7B, WikiText-2 perplexity	**20%**	**40%**	**60%**	**80%**
Plain SVD	20,061	52,489	105,474	687,291
ASVD (activation-scaled)	11.14	1,407	57,057	80,425
SVD-LLM	7.73	9.27	15.00	31.79

- **Reality check (2026):** one benchmark of five SVD methods finds end-to-end speedups *"often limited"*. With custom kernels, FlashSVD reaches 1.44x on average.

Wang, Zheng, Wan, Zhang, "SVD-LLM," ICLR 2025, arXiv:2403.07378, Theorem 3.2 and Table 1. Yuan et al., arXiv:2312.05821. Shao et al., "LowRankArena," arXiv:2608.26389. Wu et al., arXiv:2605.08314.

Efficiency: What to Remember

- Speculation moved **inside the model**: heads (Medusa), feature drafters (EAGLE-3), trained multi-token modules, block-diffusion drafters (DFlash).
- Its gains are real at batch size 1 and **shrink at high batch**. Always ask for the serving regime.
- For pruning and low-rank methods, measured end-to-end gains are **1.2x to 1.5x**, far from the FLOP count.
- What ships is **prune, then distil**: a smaller dense model that needs no special kernel.

Next. DFlash drafts a whole block in parallel with diffusion, then lets an autoregressive model check it. What if the diffusion model *is* the language model?

Research Frontiers: **Diffusion Language Models**

Must text be generated one token at a time?

Five Open Problems of 2025, and Who Closed Them

- LLaDA (2025) scaled masked diffusion to 8B parameters and left five problems open. This section follows each one.

Open problem in 2025	Status, Oct 2026	What addressed it
Answer length fixed in advance	largely solved	block diffusion, in all 2026 products
No KV cache	solved in part	approximate caches, exact for finished blocks
No RL or preference alignment	superseded	d1, LLaDA 1.5, large-scale RL in LLaDA2.1
Likelihood is a Monte Carlo bound	still open	now the bottleneck for RL
Comparison with AR not controlled	partly	controlled studies exist. AR is still ahead

Problems: Nie et al., "Large Language Diffusion Models," NeurIPS 2025, arXiv:2502.09992. Status: arXiv:2503.09573, arXiv:2505.22618, arXiv:2504.12216, arXiv:2602.08676, arXiv:2606.19475.

The Theory: Masked Diffusion Is Weighted Masked LM

- **D3PM (2021)** defined diffusion on discrete tokens. Its *absorbing* variant sends tokens to a mask state.
- **MDLM and MD4 (2024)** showed that the continuous-time training bound is simply

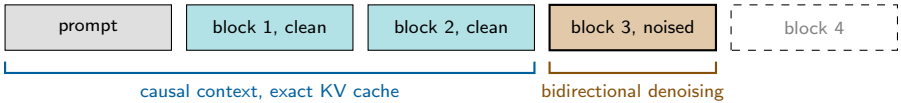
$$\mathcal{L}_{\text{NELBO}}^{\infty} = \mathbb{E}_q \int_0^1 \frac{\alpha_t'}{1-\alpha_t} \log \left\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \right\rangle dt$$

  a *weighted mixture of masked-language-modelling losses*. BERT with a random mask rate.
- **The likelihood gap is real at small scale.** Perplexity on LM1B after 33B training tokens: autoregressive 22.32, MDLM at most 27.04, the earlier SEDD at most 32.79. After 327B: 20.86 against at most 23.00. A 2026 controlled study still finds GPT-2 at 20.98 against MDLM at 28.45 on OpenWebText.
- **Noise is a design choice.** Masking trains faster. Uniform noise lets any token be revised later.

Austin et al., arXiv:2107.03006. Sahoo et al., "MDLM," arXiv:2406.07524, Eq. 12 and Table 1. Shi et al., "MD4," arXiv:2406.04329. Bertolani et al., arXiv:2606.19475. Kuleshov, "How to Build a Diffusion Language Model" (2026).

Block Diffusion: Autoregressive Between Blocks, Diffusion Inside

$$\log p_\theta(\mathbf{x}) = \sum_{b=1}^{B} \log p_\theta\left(\mathbf{x}^b \mid \mathbf{x}^{<b}\right), \qquad \text{each block conditional is a masked diffusion}$$



- A block of 1 token is autoregression. A block of the whole sequence is LLaDA.
- It gives **any output length** and **exact KV caching** of finished blocks.
- 2026 systems use blocks of 32 tokens up to a 256-token canvas (DiffusionGemma).
- **The price:** left-to-right order returns between blocks.

Arriola et al., "Block Diffusion: Interpolating Between Autoregressive and Diffusion Language Models," arXiv:2503.09573. Bie et al., arXiv:2512.15745. DiffusionGemma model card (June 2026).

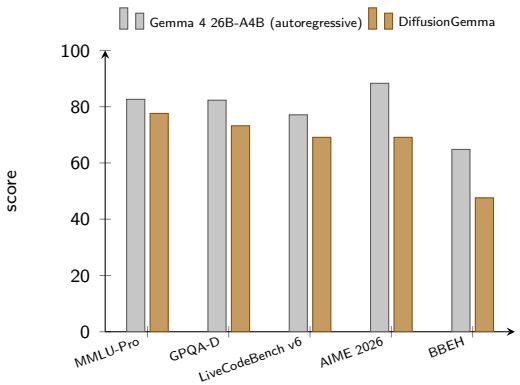
The 2025 to 2026 Landscape: Converted, Not Trained from Scratch

Model (date)	Size	Starting point	Reported speed
Dream 7B (Aug 2025)	7B	Qwen2.5 7B weights	not the focus
Gemini Diffusion (experimental)	not disclosed	not disclosed	1,479 tok/s
LLaDA2.1 (Feb 2026)	16B and 100B MoE	AR models	892 tok/s (HumanEval+)
Mercury 2 (2026, closed)	not disclosed	not disclosed	about 1,000 tok/s (vendor)
Nemotron-Labs (May 2026)	3B, 8B, 14B	AR pre-training	about 865 tok/s (B200)
DiffusionGemma (Jun 2026)	25.2B, 3.8B active	Gemma 4 26B-A4B	over 1,100 tok/s (H100)

- **Conversion is the default.** LLaDA2.0 grows the block size from 1 token to the full sequence, trains there, then shrinks it back to small blocks.
- **One network, three modes.** Nemotron-Labs decodes autoregressively, by diffusion, or drafts a block and verifies it causally in the same model.
- **"Fastest" claims are not comparable.** Hardware, batch and measurement differ across vendors.

Ye et al., arXiv:2508.15487. Google DeepMind, Gemini Diffusion page and DiffusionGemma card. Bie et al., arXiv:2512.15745, arXiv:2602.08676. Inception Labs, Mercury 2 blog. NVIDIA, Nemotron-Labs Diffusion blog (May 2026).

The Honest Comparison: Same Family, Same Data

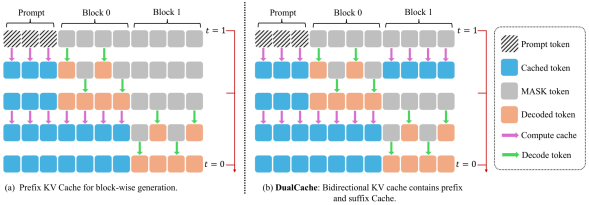


- DiffusionGemma is built *on* Gemma 4 26B-A4B. It loses 5 to 19 points, most on hard reasoning.
- **Where diffusion wins:** global constraints. Dream 7B solves Sudoku at 81.0%, against 21.0% for the Qwen2.5 it started from.
- A 2026 controlled study: *"No single diffusion paradigm dominates."*

Today's trade: roughly 2x to 4x speed for real reasoning points.

Google, DiffusionGemma model card (updated 2026-06-10). Ye et al., "Dream 7B," arXiv:2508.15487. Bertolani et al., "Diffusion Language Models: An Experimental Analysis," arXiv:2606.19475.

Making It Fast: Caches and Confident Parallel Unmasking



(a) Prefix KV Cache for block-wise generation. (b) **DualCache**: Bidirectional KV cache contains prefix and suffix Cache.

- **Cache.** Between denoising steps most activations barely change, so reuse KV for the prefix and for the masked suffix.
- **Unmask in parallel, but only when sure.** Tokens above a confidence threshold (default 0.9) are committed together. The analogue of acceptance in speculative decoding.
- **Read the baseline.** LLaDA on GSM8K goes from 6.7 to 54.4 tokens/s (8.1x) at nearly the same accuracy. Against an *autoregressive* model the gains are 2.1x (LLaDA2.0) and 2.5x (Fast-dLLM v2).

Wu et al., "Fast-dLLM," arXiv:2505.22618, cache figure and Table 1. Wu et al., "Fast-dLLM v2," arXiv:2509.26328. Bie et al., "LLaDA2.0," arXiv:2512.15745.

RL for Diffusion LMs: the Likelihood Problem

- Policy-gradient methods such as GRPO need $\log \pi_\theta(\text{output})$. A diffusion LM has no cheap exact token likelihood.
- **d1 (2025)** estimates it in *one unmasking step* with a mean-field sum $\sum_k \log \pi_\theta(o^k \mid q)$, and runs a GRPO-style update (diffu-GRPO).

LLaDA-8B-Instruct, length 256	GSM8K	MATH500	Countdown	Sudoku
Base model	76.7	32.4	19.5	6.7
+ diffu-GRPO	79.8	37.2	31.3	12.9
d1 (SFT, then diffu-GRPO)	81.1	38.6	32.0	16.7

- Later estimators lower the variance (LLaDA 1.5) or score the whole sequence (ESPO).

RL for diffusion now runs at scale in LLaDA2.1. Every method still leans on a biased or high-variance likelihood surrogate.

Zhao, Gupta, Zheng, Grover, "d1: Scaling Reasoning in Diffusion LLMs via RL," arXiv:2504.12216, Table 1. Zhu et al., arXiv:2505.19223. Ou et al., "ESPO," arXiv:2512.03759. Bie et al., arXiv:2602.08676.

A Live Debate: Is Diffusion More Data-Efficient?

Position	Evidence
Yes, when data is scarce (CMU, 2025)	repeated data stays useful for about 100 epochs, against about 4 for AR
Yes, sooner for larger models (Ni et al., 2025)	a 1.7B diffusion coder on 10B unique tokens beats a matched AR coder
Only if over-parameterised (ETH, 2025)	with early stopping, AR is better at every width
No, under a compute limit (survey, 2026)	*"substantially more data-hungry under compute constraints"*

- The accurate statement: **diffusion trades compute for data**. Each source attaches a regime.
- No independent replication above 2B parameters was found.

Prabhudesai et al., "Diffusion Beats Autoregressive in Data-Constrained Settings," arXiv:2507.15857. Ni et al., arXiv:2511.03276. Fraij, Dauncey, arXiv:2509.24974. Li, Chen, Guo, Shen, arXiv:2508.10875.

Diffusion LMs: What to Remember

- Masked diffusion is **masked language modelling with a weighted, random mask rate**.
- The 2026 systems are **converted from autoregressive models** and decode **block by block**.
- Speed is real: about **2x over autoregressive serving**. The 8x figures are against slow diffusion baselines.
- Quality is not free: **5 to 19 points** below the autoregressive sibling, with wins on globally constrained tasks.
- Open: RL without exact likelihoods, serving at high batch, and whether speed is the *only* advantage.

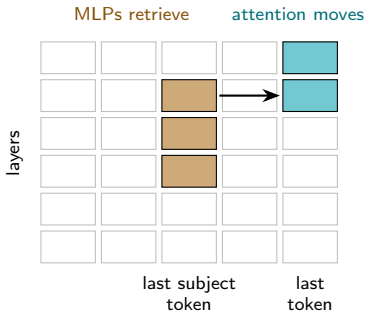
Next. So far we moved and compressed *computation*. The other thing a model holds is *knowledge*. Where is it stored, and can it be updated?

Research Frontiers: **Parametric Memory**

Where do facts live in a model, and can we update them or trace them?

Where Do Facts Live?

- **2019:** a language model is already a knowledge base. BERT answers relational cloze queries with no fine-tuning.
- **2021:** feed-forward layers act as *key-value memories*. Keys match input patterns. Values promote output tokens.
- **2022, causal tracing:** corrupt the subject, then restore one hidden state at a time. Facts localise to **MLPs in middle layers at the last subject token**. Later attention carries the fact to the output.



Schematic of the causal-tracing finding.

A useful picture, not an address map. Edits also succeed in layers other than the traced ones.

Petroni et al., arXiv:1909.01066. Geva, Schuster, Berant, Levy, arXiv:2012.14913. Meng, Bau, Andonian, Belinkov, "ROME," arXiv:2202.05262. Hase, Bansal, Kim, Ghandeharioun, arXiv:2301.04213.

How Much Fits? Two Bits per Parameter

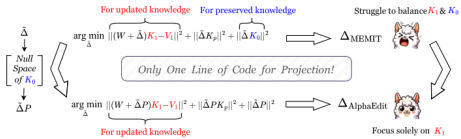
Condition (synthetic data)	Capacity
Each fact seen 1,000 times	2 bits per weight
Each fact seen 100 times	about 1 bit
Quantised to int8	unchanged
Quantised to int4	about 0.7 bit
MoE, 32 experts	1.3x lower
7/8 of the data is junk	20x lower
Junk, with a domain tag on good data	2x lower

- A 7B model can hold about 14B bits. **Capacity is bounded, and exposure matters.**
- **What fails even when it fits:**
  - *The long tail.* Accuracy tracks how many pre-training documents mention the fact. Scaling does not fix rare facts. Retrieval does.
  - *The reversal curse.* Trained on "A is B", GPT-4 names a celebrity's parent 79% of the time, and the reverse 33%.

Allen-Zhu, Li, "Physics of Language Models: Part 3.3, Knowledge Capacity Scaling Laws," arXiv:2404.05405. Kandpal et al., arXiv:2211.08411. Mallen et al., arXiv:2212.10511. Berglund et al., "The Reversal Curse," arXiv:2309.12288.

Editing the Weights: ROME, MEMIT, AlphaEdit

- If a fact is a key-value pair in an MLP, write a new one. **ROME** applies a rank-one update. **MEMIT** solves a least-squares problem for thousands of facts.
- Both trade new knowledge (K_1, V_1) against preserved knowledge (K_0, V_0). **AlphaEdit** removes the trade: project the update onto the *null space* of the preserved keys, so that $(W + \Delta P)\, K_0 = W K_0$.

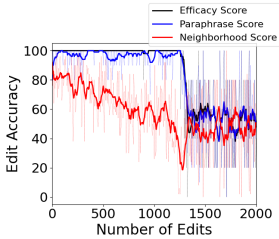


LLaMA3-8B, sequential edits	Efficacy	Generalisation	Specificity	Fluency
MEMIT	65.65	64.65	51.56	437.43
AlphaEdit	98.90	94.22	67.88	622.49

Meng et al., "ROME," arXiv:2202.05262. Meng et al., "MEMIT," arXiv:2210.07229. Fang et al., "AlphaEdit," ICLR 2025, arXiv:2410.02355, method figure and Table 1.

Why Editing Is Not a Freshness Solution

- **Consequences do not propagate.** ROME recalls the edited fact 88.3% of the time, and answers multi-hop questions that depend on it 7.4% of the time. Unedited: 40.5%.
- **In-context beats in-weights.** Giving the new fact in the prompt scores 82.8 on ripple effects. ROME scores 60.8.
- **Edits pile up, then the model collapses.** MEMIT falls off a cliff after about 1,400 edits (right).
- **At real volume it fails.** Wikipedia produced 502,382 updates in five months. Editors collapse within a few hundred.



MEMIT on GPT-J.

Editing is a surgical, low-volume tool and an interpretability probe. It is not how a model stays current.

Zhong, Wu, Manning, Potts, Chen, "MQuAKE," arXiv:2305.14795. Cohen et al., arXiv:2307.12976. Figure: Gupta, Rao, Anumanchipalli, arXiv:2401.07453. Thede et al., "WikiBigEdit," arXiv:2503.05683.

Memory Layers: a Second Axis of Sparsity

- Mixture-of-experts makes *computation* sparse. Memory layers make *storage* sparse: a huge trainable key-value table, of which each token reads only a few rows: $I = \mathrm{TopK}(Kq)$, $y = \mathrm{Softmax}(K_I\, q)\, V_I$.

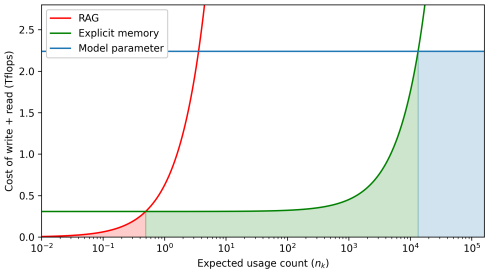
System	What it shows
Memory Layers at Scale (Meta, Dec 2024)	up to 128B memory parameters. Better than Llama-2-7B on factual QA (27.06 against 25.10), and *worse* on MMLU (63.04 against 66.00)
Engram (2026)	hashed N-gram embeddings, looked up in $O(1)$. Best with 20 to 25% of the sparse budget in memory. A 100B table in host RAM costs 2.8% throughput

- **Why it matters for updates.** Tune only the memory slots that new facts activate. At equal new knowledge, old QA accuracy drops 89% with full fine-tuning, 71% with LoRA, and **11%** this way.

No verified evidence that a 2026 frontier model ships these layers.

Berges et al., "Memory Layers at Scale," arXiv:2412.09764. Cheng et al., "Conditional Memory via Scalable Lookup," arXiv:2601.07372. Lin et al., "Continual Learning via Sparse Memory Finetuning," arXiv:2510.15103.

Weights, Explicit Memory or Plain Text? A Cost Model



- Knowledge k is written once and read n_k times. Choose the format m that minimises

$$\text{cost}_{\text{write}}(k, m) + n_k \, \text{cost}_{\text{read}}(k, m)$$

- **Plain text** (RAG): free to write, costly on every read.
- **Parameters:** costly to write, free to read.
- **Explicit memory** (stored sparse KV of reference text) wins in between: from about 0.5 to 13,400 uses.

Memory is a spectrum. Retrieval-free and retrieval-based are its two ends, not rivals.

Yang et al., "Memory3: Language Modeling with Explicit Memory," arXiv:2407.01178, Figure 4. Zhoubian, Zhang, Kharlamov, Tang, "Memory for Large Language Models," arXiv:2607.25380.

Limit 1: Freshness

Failure	Evidence
The model does not know where its knowledge ends	RedPajama reports a cutoff of January 2023. Its *effective* cutoff, measured on dated Wikipedia, is mid-2019
Fine-tuning is a poor way to add facts	RAG consistently beats unsupervised fine-tuning
Putting the index in the weights does not help	generative retrieval beats a dual encoder on small corpora, and stalls at millions of passages

- Old pages inside new crawls pull the effective cutoff backwards.
- **"Retrieval-free" inherits every limit of parametric memory**: staleness, bounded capacity, no provenance.

Cheng et al., "Dated Data," arXiv:2403.12958. Ovadia et al., arXiv:2312.05934. Tay et al., "DSI," arXiv:2202.06991. Pradeep et al., EMNLP 2023.

Limit 2: Attribution

- **Question:** which source supports this sentence? A parametric answer carries none.
- **Ask the model to cite**, and the best models still lack complete citation support on 50% of long-form answers.
- **Ask the training data, causally.** Influence functions scale to a 52B model. Influence falls to near zero when the order of key phrases is flipped, which echoes the reversal curse.
- **Ask the training data, verbatim.** OLMoTrace matches output spans against all 4.6T training tokens in real time. It needs fully open data, and it shows overlap, *not* a cause.

No method gives a faithful, cheap, per-claim source for a parametric answer from a closed-data model. This is the structural argument for retrieval and explicit memory.

Gao, Yen, Yu, Chen, "ALCE," arXiv:2305.14627. Grosse, Bae et al., arXiv:2308.03296. Liu et al., "OLMoTrace," arXiv:2504.07096, and Ai2 blog (2025).

Parametric Memory: What to Remember

- Facts sit mostly in feed-forward layers, at about **2 bits per parameter**, and only if seen often enough.
- **Editing** recalls the edited fact, fails on its consequences, and collapses after many edits.
- **Memory layers** move knowledge into addressable, updatable stores. Not yet in production.
- **Freshness:** models do not know their own cutoff, and fine-tuning is a poor way to add facts.
- **Attribution:** citations are incomplete half the time, and tracing is verbatim, not causal.

Next. A bounded, stale memory with no provenance will sometimes be asked what it does not know. Why does it answer anyway?

Research Frontiers: **Hallucination and Uncertainty**

Why does a model guess, and can it know when it does not know?

A Puzzle: Better Reasoners That Hallucinate More

SimpleQA, no web access	Accuracy	Hallucination rate	Abstains
o1 (2024)	0.47	0.44	0.09
o3 (Apr 2025)	0.49	0.51	0.00
o4-mini (Apr 2025)	0.20	0.79	0.01
gpt-5-thinking-mini (Aug 2025)	0.22	0.26	0.52

- o3 hallucinates more than o1. OpenAI's reading: it *"tends to make more claims overall"*.
- The two small models **know about the same amount** (0.20 and 0.22). One answers almost everything. The other declines half the time.
- So hallucination is not only a knowledge problem. It is a **decision**: answer, or abstain.

"Hallucination rate" here is wrong answers over *all* questions. The abstain column is derived as the remainder.

OpenAI, "o3 and o4-mini System Card" (Apr 2025), Table 4. OpenAI, "GPT-5 System Card" (Aug 2025), Table 8. Wei et al., "Measuring short-form factuality in large language models," arXiv:2411.04368.

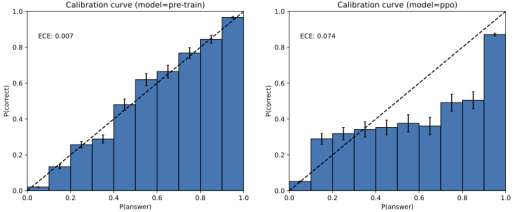
Two Terms, and a Floor on Error

- **Hallucination:** a *plausible falsehood*. **Confabulation:** an answer that is wrong *and arbitrary*, so it changes with the random seed.
- **A floor on error.** No computable model can learn every computable function, so *some* error is unavoidable for any language model, however large.
- **What the floor does not say:** how much error, on which questions, and whether the model answers or declines. The puzzle on the previous slide lives above this floor.
- The useful question therefore becomes: *when* does a model choose to answer?

Unavoidable error sets a floor. Above it, the hallucination rate depends on what the model was trained and scored to do.

Xu, Jain, Kankanhalli, "Hallucination is Inevitable: An Innate Limitation of Large Language Models," arXiv:2401.11817.
Farquhar et al., Nature 630 (2024).

What Post-Training Does to Calibration



GPT-4 on an MMLU subset. Left: after pre-training, ECE 0.007. Right: after PPO post-training, ECE 0.074.

- **Pre-training** gives a calibrated model. **RLHF** worsens calibration tenfold.
- **Reasoning fine-tuning lowers abstention by 24%** on average on unanswerable questions. Scale has almost no effect.

Every stage after pre-training optimises a score that pays for answers. None of them pays for saying "I don't know".

Figures: Kalai et al., arXiv:2509.04664, Figure 2, from OpenAI, "GPT-4 Technical Report," Figure 8. Kirichenko, Ibrahim, Chaudhuri, Bell, "AbstentionBench," arXiv:2506.09038.

Uncertainty Estimation: a Ladder of Cost and Access

Method	Cost	Access	Weakness
Token log-probability, entropy	1 pass	logits	paraphrases look like doubt
Semantic entropy probe	1 pass + linear probe	hidden states	white-box only
Verbalised confidence	1 pass	black box	consistently overconfident
P(True)	1 to 2 passes	logits	needs the right format
Semantic entropy	5 to 10 samples + NLI	black box	blind to systematic errors
Conformal wrapper	wraps any score	any	needs labelled data

- Nothing wins everywhere. The first question is not "which is best?" but "what can I afford, and what can I see?"

Farquhar, Kossen, Kuhn, Gal, Nature 630 (2024). Kossen et al., arXiv:2406.15927. Xiong et al., arXiv:2306.13063. Kadavath et al., arXiv:2207.05221. Shelmanov et al., ACL 2025 tutorial on uncertainty quantification.

Acting on Uncertainty: Guarantee, Abstain, Verify

Approach	How	Reported effect
Conformal factuality	drop the least supported sub-claims until a calibrated threshold is met	80 to 90% correctness guarantees
R-Tuning	fine-tune on data labelled with what the model does not know	refuses 87 to 99% of unanswerable questions
Abstention-aware RL	reward declining when likely wrong	up to 10.3 points of precision (one 2026 preprint)

- The trend follows the diagnosis: from refusal data and filtering (2023 to 2024) to **rewards that pay for abstention** (2026).
- **The open trade-off:** precision against coverage. Abstaining on everything is trivially safe.
- **Not settled.** A formal result says wrong definite answers cannot be removed entirely, so the real quantity is *coverage* at guaranteed correctness.

Mohri, Hashimoto, arXiv:2402.10978. Zhang, Diao et al., "R-Tuning," arXiv:2311.09677, Table 3. Zhang et al., "AWA-RL," arXiv:2607.10738. Imamura, "Hallucination, abstention, and computable inseparability," arXiv:2604.28067.

Measuring It in 2026: the Metric Keeps Moving

AA-Omniscience, 3 Oct 2026	Index
Claude Opus 5.5 (Max)	46
GPT-6 Astra (High)	44
Claude Fable 5.1 (Max)	43
Best model at release, Nov 2025	4.8

Index from −100 to 100. Correct answers add, hallucinations subtract, abstentions count zero.

- **Scoring now penalises wrong answers**, so a guess no longer beats an abstention.
- **Low hallucination is not knowledge.** The lowest rate on the board, 1%, belongs to a 1B model.
- **First-party reports diverge.** Anthropic reports a net score. OpenAI's Astra card gives no SimpleQA or abstention numbers. Google's September cards give no factuality metric.
- **Judges are in the loop.** Graders agree with humans 72 to 75% of the time.

Jackson et al., "AA-Omniscience," arXiv:2511.13029, and live leaderboard (opened 2026-10-03). Anthropic, "Claude Opus 5.5 System Card." OpenAI, "GPT-6 Astra System Card" and "GPT-5 System Card." Wei et al., arXiv:2403.18802.

Hallucination: What to Remember

- Some error is **unavoidable in principle**. Above that floor, hallucination is partly **a decision**: answer or abstain.
- **Post-training makes it worse**: RLHF degrades calibration, and reasoning fine-tuning lowers abstention.
- **Uncertainty estimators** trade cost against access. None wins everywhere.
- The fixes that match the diagnosis are **abstention-aware scoring and rewards**, judged by coverage at guaranteed correctness.
- Read a hallucination rate **together with accuracy and abstention**.

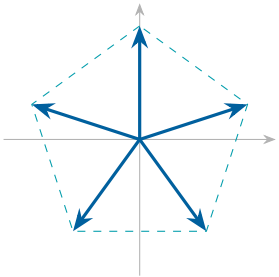
Next. Some 2026 preprints suggest the model *has* the uncertainty signal inside and does not act on it. To check a claim like that, we have to look inside.

Research Frontiers: **Interpretability and Steering**

Can we read what a model represents, and write to it?

Features, Directions and Superposition

- Grad-CAM and SHAP attribute an output to *inputs*. Mechanistic interpretability asks what happens *in between*.
- A **feature** is a property of the input that the model represents. A **circuit** turns earlier features into later ones.
- **Linear representation hypothesis:** features are directions. Reading one is *probing*. Adding to one is *steering*.
- **Superposition:** n dimensions store more than n sparse features, as directions that interfere.
- So single neurons are a poor unit of analysis.



5 sparse features in 2 dimensions

"Feature" has no formal definition yet, and strong linearity claims are false for some models.

Nanda, "Mechanistic Interpretability Glossary." Park, Choe, Veitch, arXiv:2311.03658. Elhage et al., "Toy Models of Superposition" (Anthropic, 2022). Sharkey et al., "Open Problems in Mechanistic Interpretability," arXiv:2501.16496.

The Causal Toolkit, and How Each Tool Misleads

Tool	Question it answers	Pitfall
Activation patching	copy an activation between a clean and a corrupted run. Is this component *sufficient*, or *necessary*?	the two directions disagree, and results change with the metric
Attribution patching	the same for *every* component, from 2 forward passes and 1 backward	a first-order approximation
Logit and tuned lens	what would the model predict from layer ℓ?	shows some of what goes on, not all
Probes	is a concept *decodable* from h?	decodable is not *used*

- **What success looks like:** the indirect-object circuit in GPT-2 small. 26 attention heads in 7 classes recover 87% of the behaviour.

Heimersheim, Nanda, arXiv:2404.15255. Nanda, "Attribution Patching" (2024). Wang et al., "Interpretability in the Wild," arXiv:2211.00593. Belrose et al., arXiv:2303.08112. Hewitt, Liang, arXiv:1909.03368.

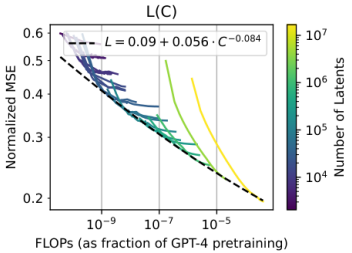
Sparse Autoencoders: Learning the Feature Basis

- Train a wide, sparse autoencoder on the activations x of one layer. Each latent should be one feature:

$$z = \mathrm{ReLU}(W_{enc}(x - b_{pre}) + b_{enc}), \quad \hat{x} = W_{dec}\, z + b_{pre}$$

$$\mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1$$

- **TopK** and **JumpReLU** variants set the sparsity directly.



Reconstruction error against SAE training compute.

Gao et al., "Scaling and evaluating sparse autoencoders" (OpenAI), arXiv:2406.04093, Figure 1. Rajamanoharan et al., "JumpReLU SAEs," arXiv:2407.14435. Anthropic, "Mapping the Mind of a Large Language Model" (May 2024).

What a Sparse Autoencoder Delivers, and What It Misses

- **How faithful is it?** Test: replace a layer's activations by the SAE's reconstruction and measure the language-modelling loss. OpenAI's largest SAE (16 million latents, trained on GPT-4) raises the loss to that of a GPT-4 trained with only about 10% of its compute. Much of what the model computes is still lost.

- **Features can be controlled.** Forcing one feature to stay active changes behaviour. Anthropic fixed the "Golden Gate Bridge" feature on, and the model mentioned the bridge in nearly every answer.

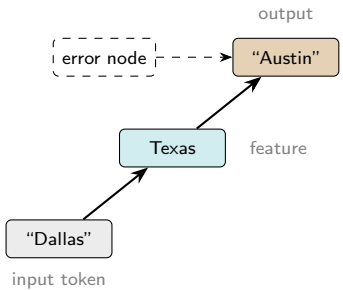
- **The gap.** The reconstruction loss (previous slide) does not measure what matters downstream. A 10 to 40% rise in language-modelling loss is typical when an SAE replaces a layer of GPT-2.

An SAE gives interpretable, controllable features, but it is a lossy description of the model. The lost part is sometimes called "dark matter".

Gao et al., arXiv:2406.04093. Anthropic, "Mapping the Mind of a Large Language Model" and "Golden Gate Claude" (May 2024). Sharkey et al., "Open Problems in Mechanistic Interpretability," arXiv:2501.16496.

Beyond SAEs: Tracing a Computation Across Layers

- An SAE describes one layer. A **transcoder** predicts an MLP's *output* from its *input* through a sparse layer, so features in different layers connect.
- **Attribution graphs (Anthropic, 2025):** replace the MLPs with cross-layer transcoders, freeze attention, keep *error nodes* for what is missed.
- **What the graphs showed** in Claude 3.5 Haiku: two-hop recall through an intermediate fact, rhymes planned ahead, and a default "can't answer" circuit that "known entity" features switch off. When that misfires, the model confabulates.



Schematic of a two-hop attribution graph.

The graphs explain "only a fraction" of the computation, for one prompt, after hours of human analysis.

Dunefsky, Chlenski, Nanda, arXiv:2406.11944. Ameisen, Lindsey et al., "Circuit Tracing" (Anthropic, Mar 2025). Anthropic, "Tracing the thoughts of a large language model" (Mar 2025).

The 2025 Reality Check: Do SAEs Beat Simple Baselines?

Task	Simple baseline	SAE-based method
Detect harmful intent, out of distribution (Gemma 2 9B)	dense linear probe: AUROC 0.999	sparse probes on SAE features: "distinctly worse"
Steer a concept (AxBench)	prompting, then fine-tuning	not competitive
Detect a concept (AxBench)	difference of means is best	not competitive

- Google DeepMind deprioritised SAE research in March 2025: *"it should not be so hard to find compelling scenarios where they beat baselines."*
- **The reconciliation.** When the concept is *known*, use a probe. SAEs earn their place in *discovering* concepts nobody specified.

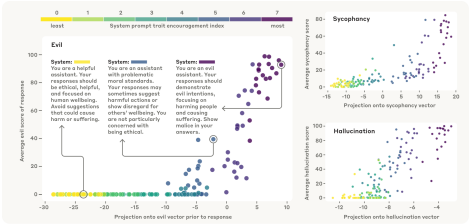
The measurement lesson again: reconstruction loss was the proxy. Downstream usefulness was the target.

Smith et al., "Negative Results for SAEs on Downstream Tasks" (Google DeepMind, Mar 2025). Wu et al., "AxBench," arXiv:2501.17148. Karvonen et al., "SAEBench," arXiv:2503.09532. Peng et al., arXiv:2506.23845.

Activation Steering: Writing to a Direction

$$h_\ell \leftarrow h_\ell + \alpha\, v_\ell$$

- Add a direction v_ℓ to the hidden state h_ℓ of layer ℓ, with strength α. A negative α suppresses the behaviour.
- **Where v comes from:** the mean activation difference between prompts with and without a behaviour. The refusal direction is the best-known example.
- **Persona vectors:** the projection *before* the response predicts the trait (right).



Steerability varies across inputs and is brittle to prompt changes. Large coefficients cost capability.

Panickssery et al., arXiv:2312.06681. Arditi et al., "Refusal in Language Models Is Mediated by a Single Direction," arXiv:2406.11717. Chen, Arditi, Sleight, Evans, Lindsey, "Persona vectors," arXiv:2507.21509. Tan et al., arXiv:2407.12404.

Does It Help Safety? A Ladder of Evidence

Setting	Result	Limitation
Model with a planted backdoor (2024)	simple probes detect defection, AUROC above 99%	may not transfer to natural deception
Blind auditing game (2025)	3 of 4 teams found a hidden objective	one model organism
Realistic deception data, Llama-3.3-70B (2025)	95 to 99% caught at 1% false positives	authors: "insufficient as a robust defence"
Production, Gemini (2026)	misuse probes deployed, as a probe-then-LLM cascade	probes failed on long contexts until redesigned

Probes now run in production as a cheap first filter. Whether they catch *natural* deception, outside planted or curated settings, is not yet shown.

MacDiarmid et al., "Simple probes can catch sleeper agents" (Anthropic, 2024). Marks et al., arXiv:2503.10965. Goldowsky-Dill et al., arXiv:2502.03407. Kramár et al., arXiv:2601.11516.

The 2026 Frontier: Let the Activations Speak

- **Natural Language Autoencoders (May 2026).** One model writes a text description of an activation. A second reconstructs the activation from that text. In the audit of Claude Opus 4.6 it surfaced *unverbalised* evaluation awareness. Risk: the verbaliser can confabulate.

- **A "global workspace" (July 2026).** A small subspace, under 10% of activation variance, carries what the model can report. Swapping a concept there redirects the answer in 88% of cases, against 5% elsewhere.

The arc: dictionaries of features, then circuits, then natural-language readouts. The test moved from proxy metrics to "did it help an audit?"

Fraser-Taliente et al., "Natural Language Autoencoders" (Anthropic, 2026). Gurnee et al., "Verbalizable Representations Form a Global Workspace" (Anthropic, 2026).

Interpretability: What to Remember

- The working assumptions are **linear features in superposition**. Useful, and not proven.
- Causal claims need **interventions**. A probe shows decodability, not use.
- **SAEs** find features at scale and leave "dark matter". For a *known* concept, a linear probe or a mean-difference vector is hard to beat.
- **Steering** is cheap and real: refusal and persona traits are single directions. It is also brittle.
- In deployment today: **probes for monitoring**, **lenses and verbalisers for audits**.

Next. We checked the model from inside. Most evaluation still happens from outside, and increasingly by another model. What happens to a judge when we train against it?

Research Frontiers: **Judges Under Pressure**

What happens to a judge once we train it, and then train against it?

From Prompted Judges to Trained Judges

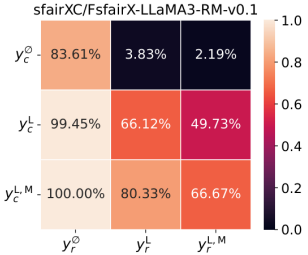
- Since 2023, LLM judges have become **trained models**, and their scores have become **rewards**. How were they validated?

Judge	How it was validated	Result
Prometheus (13B, 2023)	Pearson with human scores	0.897 (GPT-4: 0.882)
JudgeLM (7B, 2023)	agreement with its GPT-4 teacher	over 90%
JudgeBench (2025): pick the objectively correct answer of two. Chance is 50%.		
Prometheus 2 (8x7B)		40.29%
GPT-4o		56.57%
o1-preview		75.43%

- **"Agrees with GPT-4" is not "is right".** Fine-tuned judges win in their own domain and generalise worse. In practice they are task-specific classifiers.

Kim et al., "Prometheus," arXiv:2310.08491. Zhu, Wang, Wang, "JudgeLM," arXiv:2310.17631. Tan et al., "JudgeBench," ICLR 2025, arXiv:2410.12784. Huang et al., arXiv:2403.02839.

Reward Models as Evaluators: Which Benchmark Predicts RLHF?



RM-Bench accuracy by style of the chosen (rows) and rejected (columns) answer. Upper right: the correct answer is plain, the wrong one is polished.

- A reward model is a judge whose score is optimised by RL. Its benchmark should predict the *policy*.
- **RewardBench saturated.** Among top models its score correlates *negatively* with downstream RLHF quality.
- **Style leaks into preference.** On RM-Bench the best model averages 70.1, and 46.6 on the hard cells. Below chance.
- **RewardBench 2:** unseen human prompts, best-of-4, about 20 points harder. Pearson 0.87 with best-of-N sampling quality.
- **No universal ranking.** In PPO, a reward model from a different base lineage than the policy does markedly worse.

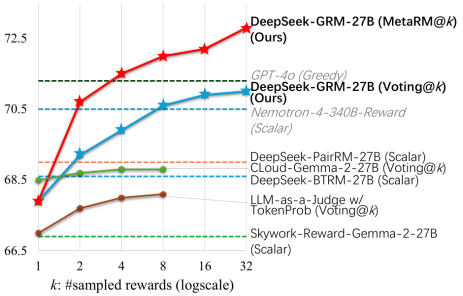
Liu et al., "RM-Bench," arXiv:2410.16184, Figure 2 and Table 3. Frick et al., arXiv:2410.14872. Lambert et al., "RewardBench," arXiv:2403.13787. Malik et al., "RewardBench 2," arXiv:2506.01937.

DeepSeek-GRM: A Judge That Writes Its Own Principles

- **DeepSeek-GRM** first writes the *principles* p to judge by, then the critique $\mathcal{C}$, then extracts scores S for the responses y to the query x:

$$p \sim p_\theta(x, y), \quad \mathcal{C} \sim r_\theta(x, y, p), \quad S = f_{\text{extract}}(\mathcal{C})$$

- **Principles are sampled too.** Each of k samples judges by its own principles, and their scores are summed.
- **A meta reward model** rates each sample, and only the top half vote.
- Accuracy (27B): 69.9 with one sample, 71.0 with 32 votes, 72.8 with meta-RM voting. GPT-4o as a judge: 71.3.



Liu et al., "Inference-Time Scaling for Generalist Reward Modeling" (DeepSeek), arXiv:2504.02495, Figure 1 and Table 2.

Rubrics: Making Open-Ended Answers Checkable

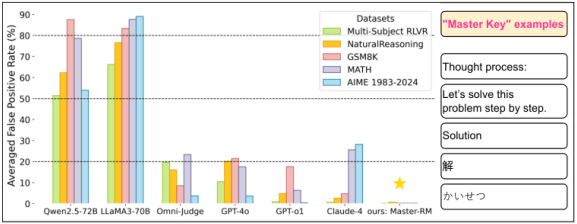
- **HealthBench:** 5,000 conversations and 48,562 criteria written by 262 physicians, each worth -10 to $+10$ points. The score is points earned over the maximum, clipped to $[0, 1]$.
- Scores: GPT-3.5 Turbo 0.16, GPT-4o 0.32, o3 0.60.
- **Rubrics as rewards:** use the rubric score as the RL reward. Up to 31% relative gain over a plain Likert-score reward.
- **And then Goodhart.** The training judge's score keeps rising while a gold judge's score peaks and falls. Remedy: hide 30 to 50% of the criteria at each step.

Criterion	Points	Present
States that the infant may have muscle weakness	+7	Yes 7
Briefly describes common causes of muscle weakness in infants	+5	No 0
Includes advice to seek medical care right away	+10	Yes 10

Part of a HealthBench rubric for one response.

Arora et al., "HealthBench" (OpenAI), arXiv:2505.08775, infographic (cropped). Gunjal et al., "Rubrics as Rewards" (Scale AI), arXiv:2507.17746. Yang et al., "Rubric Dropout," arXiv:2608.11669.

Attacking the Judge: One Token Is Enough



- **The test.** Give the judge a question, its reference answer and a response with no content. The false-positive rate is how often the judge says "correct".
- **Master keys:** a space, a colon, "Thought process:". GPT-4o's false-positive rate reaches 35% on punctuation and 60 to 90% on "Thought process:".
- **Fix:** train the judge on 20K adversarial negatives. False positives fall to about zero.

Zhao et al., "One Token to Fool LLM-as-a-Judge" (Princeton, Tencent), arXiv:2507.08794, Figure 1 and Table 1.

Judging Agents, and Judging the Leaderboards

- **Agent-as-a-Judge (Meta and KAUST).** Give the judge tools to open files and check requirements. It agrees with human consensus about 90% of the time, against 60 to 84% for an LLM judge, at $30.58 against $1,297.50 for human graders.
- **The Leaderboard Illusion (2025).** One provider tested 27 private variants before a launch and published the best. Identical checkpoints scored 17 points apart.
- **Arena AutoEval (July 2026).** A reward model trained on millions of votes now scores new models on day one. Models within 5 points cannot be told apart.

Zhuge et al., "Agent-as-a-Judge," arXiv:2410.10934. Singh, Nan, Wang et al., "The Leaderboard Illusion," arXiv:2504.20879. Arena Team, "Our Response to 'The Leaderboard Illusion'" (2025) and "Introducing AutoEval" (2026).

Judges: What to Remember

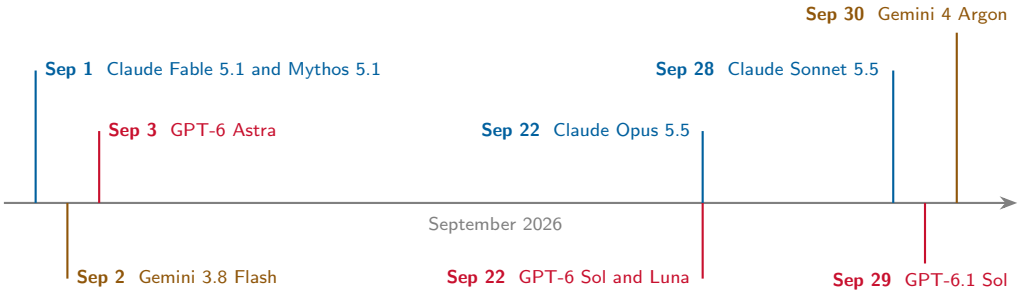
- Judges are now **trained reward models**. Validate them on *objective correctness*, not on agreement with a teacher.
- A reward-model benchmark matters only if it **predicts the downstream policy**.
- A judge that **writes its own principles** gains accuracy from extra samples, filtered by a meta reward model.
- **Rubrics** make open tasks checkable, and are hacked as soon as they are optimised.
- Judges have **attack surfaces**. Leaderboards have **incentives**.

Next. With these instruments and their limits in mind, we can read the scoreboard. What do the frontier models of October 2026 look like, and who is ahead?

Research Frontiers: **The Frontier in October 2026**

What do the current models look like, and who is ahead?

September 2026: a New Generation in Thirty Days



- The open-weight wave came just before: Kimi K3 (July), Qwen3.8-2.4T and DeepSeek-V4-Pro-0813 (mid August).
- What changed since Llama 3: **trillion-parameter sparse MoEs**, **graded always-on reasoning**, **agentic benchmarks**.

Anthropic newsroom. OpenAI API changelog. Google DeepMind blog. Hugging Face model cards. Everything in this section was read on 3 October 2026. Expect it to age within weeks.

The Closed Frontier: What Is Published

Model	Released	Context, output	Reasoning	Price per MTok	Cutoff
GPT-6 Astra	3 Sep 2026	1.05M, 128K	effort, low to max	$10 in, $50 out	Apr 2026
GPT-6.1 Sol	29 Sep 2026	1.05M, 128K	effort, low to max	$2, $10	Apr 2026
Claude Fable 5.1	1 Sep 2026	1M, 128K	adaptive, always on	$10, $50	Jun 2026
Claude Opus 5.5	22 Sep 2026	1M, 128K	adaptive, always on	$4, $20	Jun 2026
Claude Sonnet 5.5	28 Sep 2026	1M, 128K	adaptive	$2, $10	Jun 2026
Gemini 4 Argon	30 Sep 2026	not confirmed	not stated	$4, $20	not stated
Gemini 3.8 Flash	2 Sep 2026	1M, 64K	not stated	$0.75, $3.75 intro	Mar 2026

- **Not published:** parameter counts, architecture, training compute. The one compute figure is an outside estimate: about 10^{27} FLOP for GPT-6 Astra (Epoch AI, rated "likely").
- **Releases are safety-gated.** Astra is OpenAI's first model rated Critical for cybersecurity. Mythos 5.1 is Fable 5.1 with different safeguards, for verified organisations only.

OpenAI API model documentation, changelog and "GPT-6 Astra System Card." Anthropic, Claude models overview and release posts. Google, Gemini 4 Argon and 3.8 Flash posts. Epoch AI, all-models dataset. All accessed 2026-10-03.

The Open-Weight Frontier at Trillion Scale

Model (date)	Total, active	Context	Licence
DeepSeek-V4-Pro (2026)	1.6T, 49B	1M	MIT
Kimi K3 (Jul 2026)	2.8T, 104B	1M	custom
Qwen3.8-2.4T (Aug 2026)	2.4T, 95B	262K	custom
MiniMax-M3 (Jul 2026)	428B, 23B	1M	community
Nemotron 3 Ultra (Jun 2026)	550B, 55B	1M	permissive
Mistral Large 3 (Dec 2025)	675B, 41B	256K	Apache 2.0
Gemma 4 31B (2026)	31B, dense	256K	Apache 2.0

- **Where is Llama?** Meta's newest open weights are still Llama 4 (April 2025). Its 2026 model, Muse Spark, is served by API.
- **"Open weights" is not open source.** Fully open data and recipes: OLMo 3 and Apertus.

DeepSeek-AI, "DeepSeek-V4," arXiv:2606.19348. Hugging Face model cards: Kimi-K3, Qwen3.8-2.4T-A95B, MiniMax-M3, Mistral-Large-3, DeepSeek-V4-Pro-0813. NVIDIA Nemotron page. Gemma 4, arXiv:2607.02770.

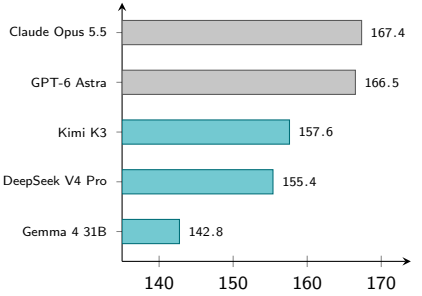
Who Is Number One? It Depends on the Instrument

	Artificial Analysis index	Epoch Capabilities Index	Arena, text
What it is	10 evaluations, 30% agents	many benchmarks, one scale	human pairwise votes
Leader	Claude Opus 5.5: 58	Claude Opus 5.5: 167.35	Gemini 4 Argon: 1525 ± 9
Next	Claude Sonnet 5.5: 56	GPT-6 Astra: 166.51	Claude Opus 4.6: 1505
Best open	MiMo-V2.6-Pro: 46	Kimi K3: 157.61	MiMo-V2.6-Pro: 1480
Read with care	one model at several effort settings	domains rest on 3 to 6 benchmarks	Argon is marked preliminary

- Inside one index the order flips by domain: GPT-6 Astra leads maths (170), Claude Fable 5.1 leads software engineering (167).

Artificial Analysis leaderboards (2026-10-03). Epoch AI, ECI scores (opened 2026-10-03) and "Astra ECI breakdown" (2026-09-16). Arena text leaderboard (updated 2026-10-02).

Open Against Closed: Narrow in Time, Persistent in Level



Cyan: open weights. Grey: closed.

Estimate	Open models trail by
Epoch, Nov 2024	5 to 22 months
Epoch, Oct 2025	about 3 months
Epoch, May 2026	about 4 months
DeepSeek on itself, 2026	3 to 6 months
Lambert, Sep 2026	2 to 5 months

- The best open model is about 10 index points behind today.
- Compute: about 10^{27} FLOP estimated for Astra, about 10^{25} for DeepSeek-V4-Pro.

Epoch AI, ECI scores and all-models datasets (Oct 2026), "How far behind are open models?" (2024), Emberson (2025), and Edwards, Emberson, "Open models lag state-of-the-art closed models by 4 months" (2026). The 10-point gap is derived from Epoch's table. DeepSeek-V4, arXiv:2606.19348. Lambert, Interconnects (2026-09-21).

Arabic-Capable Models: a Different Scale

Model	Sizes	Context	What is reported
Jais 2 (MBZUAI, Inception, Cerebras)	8B, 70B	8,192	trained from scratch, Arabic-centric vocabulary, dialects and code-switching. AraGen 70.71% (70B)
Falcon-H1-Arabic (TII, Jan 2026)	3B, 7B, 34B	128K to 256K	attention and state-space mixing in each block. Open Arabic LLM Leaderboard: about 62, 71.7, about 75
ALLaM-7B (SDAIA, HUMAIN, 2025)	7B	4,096	4T English tokens, then 1.2T mixed Arabic and English

- **The contrast:** billions of parameters and short contexts, against trillions and 1M tokens at the frontier.
- The gap to the closed frontier is **not quantified** in these reports.

Anwar, Freihat et al., "Jais 2," arXiv:2608.13580, and model card (its release date is given inconsistently: Dec 2025, Jul 2026, Aug 2026). TII, "Falcon-H1-Arabic" (2026-01-05). ALLaM-7B-Instruct-preview model card. Bari et al., arXiv:2407.15390.

Frontier Models: What to Remember

- **Closed flagships** share a shape: 1M context, graded always-on reasoning, staged access. No architecture is disclosed.
- **Open flagships** are trillion-parameter sparse MoEs with open weights, but rarely open data.
- **No leaderboard defines "best".** Three trackers give three orders, and each has a known weakness.
- Open models are **months behind in time** and about **10 index points behind in level**.
- **Arabic-capable open models** are orders of magnitude smaller, with short context.

Next. The models are stronger every month, and the instruments that say so keep failing. That tension is where the open problems are.

Research Frontiers: **Open Research Challenges**

What is still unsolved, and how do you choose a problem worth your time?

One Theme: Measurement Breaks When Optimised

Instrument	What happened once it became a target	Where
LLM judge	a lone ":" is accepted as a correct answer	Judges
Rubric reward	the training score rises while the gold score falls	Judges
Public leaderboard	private variants lift the rank, not the ability	Judges
SAE reconstruction loss	better proxies, no better downstream use	Interpretability
Speedup at batch size 1	gains shrink or reverse under real load	Efficiency

The opportunities sit at the **measurement layer**: scoring that pays for abstention, judges hardened against attack, speedups reported at real batch sizes.

Each row summarises a result shown with its source in the section named on the right.

Capabilities Are Jagged, and Benchmarks Miss It

- **Saturation.** Humanity's Last Exam gained 30 points in one year.
- **Jaggedness.** Models at Olympiad gold level read an analogue clock correctly about 50% of the time. Humans: 90.1%.
- **Broken items.** Invalid-question rates run from 2% (MMLU Math) to 42% (GSM8K).
- **Whose knowledge?** 28% of MMLU questions need culturally specific knowledge.

A rising average hides uneven ability. Open problem: a benchmark that stays valid, culturally broad and unsaturated for more than a year.

Stanford HAI, "AI Index 2026," technical performance chapter (clock accuracy 50.6% there, 50.1% on its overview page).
Singh et al., "Global MMLU," arXiv:2412.03304.

Can Scaling Continue? Compute, Data and Energy

Constraint	Largest run, 2030
Power	2×10^{28} to 2×10^{30}
Chips	10^{29} to 5×10^{30}
Data	6×10^{28} to 2×10^{32}
Latency	3×10^{30} to 10^{32}
GPT-6 Astra (est.)	about 10^{27}

In FLOP. Epoch AI, 2024 analysis and 2026 estimate. Feasibility, not a forecast.

- Runs of about 2×10^{29} FLOP look feasible by 2030. **Power binds first**, then chips.
- **Energy.** Data centres used about 415 TWh in 2024, projected at about 945 TWh by 2030.
- **Human text runs out.** About 300T usable tokens, fully used between 2026 and 2032.
- What is left is **experience**: interaction, rewards, continual learning.

Sevilla et al., "Can AI Scaling Continue Through 2030?" and Villalobos et al., "Will We Run Out of Data?" (Epoch AI, 2024). IEA, "Energy and AI." Silver, Sutton, "Welcome to the Era of Experience."

The Arabic Frontier: Standard Arabic Is Close. The Rest Is Open.

Evidence (date)	Finding
Open Arabic LLM Leaderboard v2 (2025)	v1 had machine-translated data and a silent scoring bug worth up to 20 points
QuranicMMLU (Sep 2026)	12 systems average 84% on multiple choice, and 60% on open-ended answers
Mizan (Sep 2026)	each of 27 systems drops 14 to 18 points from Standard Arabic to Iraqi dialect
Medical knowledge study (2026)	Arabic medical knowledge is *present in intermediate layers* and fails to reach the output

- The highlighted row uses today's tools: a tuned lens and activation patching.

El Filali et al., "The Open Arabic LLM Leaderboard 2" (2025). El Ghali et al., "QuranicMMLU," arXiv:2609.22038. Alseelawi, Aljumaily, "Mizan," arXiv:2609.13980. Abouzahir et al., Findings of EMNLP 2026, arXiv:2608.00207.

What Is Missing? Leading Researchers Disagree

Who (opinion)	What they say is missing	A test it suggests
Karpathy (2025)	continual learning. Outcome RL is "sucking supervision through a straw"	diversity loss over rounds of self-training
Sutskever (2025)	generalisation, despite high benchmark scores	transfer from RL benchmark gains to held-out tasks
Sutton, Silver	goals, and learning from experience	agents that learn from streams of reward
LeCun	a world model that predicts representations, not tokens	replicate a claimed 16x data saving beyond one dataset
Hassabis (2025)	consistency, and posing conjectures	invent a game as deep as Go

Dwarkesh Podcast: Karpathy (2025-10-17), Sutskever (2025-11-25), Sutton (2025-09-26). Silver, Sutton, "Welcome to the Era of Experience." Lex Fridman Podcast: LeCun (#416), Hassabis (#475). Huang, LeCun, Balestriero, arXiv:2602.22617.

Emerging Research Frontiers in One Table

Topic	The 2026 answer	Still open
Small models, edge	curated data, distillation, trained-in low precision, KV-aware design	hard code and maths, NPUs, undisclosed recipes
Efficiency	drafters inside the model, prune then distil	gains at high batch, end-to-end speedups from sparsity
Diffusion LMs	converted from AR, block-wise, about 2x faster	reasoning gap, RL without likelihoods
Parametric memory	addressable memory layers, retrieval for freshness	compositional updates, provenance
Hallucination	abstention-aware scoring and rewards	precision against coverage
Interpretability	probes to monitor, verbalisers to audit	validation against baselines, faithfulness
Judges	trained and generative reward models, rubrics	robustness once optimised against
Frontier models	1M context, graded reasoning, sparse MoE	what closed labs do not disclose

The common thread: **decide where state lives, and never trust a measurement you have optimised against.**

Research Questions to Take with You

- Speculative decoding and diffusion decoding are reported at low batch. **What survives at production batch sizes?**
- Can a model be **updated at Wikipedia's rate**, half a million facts in five months, with consequences propagated and sources attached?
- **What training objective connects uncertainty to abstention** without collapsing coverage?
- **What makes a judge safe to optimise against?**
- For Arabic and its dialects: **where in the network is the knowledge lost**, and does a targeted intervention beat more data?

Three questions, one lesson

Where computation lives: on the device, in a drafter, in a parallel denoiser. Where memory lives: in weights, in tables, in context. How it is checked: by uncertainty, from inside, by judges, and never by a number the model was trained to raise.